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A Bayesian framework for anomaly detection in scientific data based on Benford's law

Highlights

- SBBE is a Bayesian tool for assessing data conformity with Benford's law
- SBBE ranks datasets according to their curation standards
- Low Benfordness identifies various types of data quality issues
- SBBE's estimates are comparable across dataset sizes and quantify uncertainty

Authors

Uday Abu-Shehab, Matthias Welsch, Johannes Kirchmair

Correspondence

johannes.kirchmair@univie.ac.at

In brief

Simulation-based Benfordness estimation (SBBE) quantifies how closely a dataset conforms to Benford's law, the characteristic distribution of the first significant digits observed in many naturally occurring datasets and widely used in data auditing. Applicable across scientific domains, SBBE provides uncertainty estimates and enables meaningful comparisons across datasets of different sizes. When applied to bioactivity data, SBBE ranks databases by curation quality and flags anomalous subsets for which targeted expert review is likely to yield the greatest benefit.

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A Bayesian framework for anomaly detection in scientific data based on Benford's law

Uday Abu-Shehab,[1,2,4] Matthias Welsch,[1,2,3,4] and Johannes Kirchmair[1,3,5,*]
[1]Department of Pharmaceutical Sciences, Faculty of Life Sciences, University of Vienna, Josef-Holaubek-Platz 2, 1090 Vienna, Austria
[2]Vienna Doctoral School of Pharmaceutical, Nutritional and Sport Sciences (PhaNuSpo), University of Vienna, Josef-Holaubek-Platz 2, 1090 Vienna, Austria
[3]Christian Doppler Laboratory for Molecular Informatics in the Biosciences, Department of Pharmaceutical Sciences, University of Vienna, Josef-Holaubek-Platz 2, 1090 Vienna, Austria
[4]These authors contributed equally
[5]Lead contact
*Correspondence: johannes.kirchmair@univie.ac.at
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THE BIGGER PICTURE Trusted, high-quality data are fundamental to modern scientific research, yet assessing data quality remains a challenging and only partially solved problem. Most automated approaches depend on prior knowledge of the specific types of errors or inconsistencies to be detected. In contrast, one domain-independent indicator requires no such assumptions: Benford's law, the characteristic distribution of first significant digits observed in many naturally occurring datasets. According to Benford's law, numbers beginning with the digit "1" occur more frequently than those beginning with "2", which in turn occur more frequently than those beginning with "3", and so on. Deviations from this pattern can indicate unusual aspects of how data were generated, reported, recorded, or curated. However, existing measures of Benford conformity often perform poorly on large datasets and yield scores that cannot be compared directly across datasets of different sizes.

We introduce simulation-based Benfordness estimation (SBBE), a method that estimates the proportion of a dataset that conforms to Benford's law. SBBE provides uncertainty estimates and produces scores that are directly comparable across datasets regardless of size. Applied to bioactivity data, SBBE ranks databases in accordance with their curation standards and prioritizes a small subset of assay datasets for expert review. Examination of these prioritized datasets revealed diverse sources of irregularity, ranging from reporting practices, such as rounding measurements on a logarithmic scale, to genuine curation errors, including the collapse of value ranges (e.g., 100–1,000) into a single reported value. Because SBBE relies solely on the numerical values themselves, it is inherently domain agnostic and offers a rapid, versatile approach for detecting data irregularities across scientific disciplines.

SUMMARY

Ensuring data integrity is critical in modern data-intensive science, motivating the development of scalable methods for anomaly detection across datasets. Benford's law, which describes a characteristic distribution of leading digits in naturally occurring data, offers a domain-agnostic signal of irregularities, but its practical use is limited by difficulties in quantifying adherence and uncertainty, especially for datasets of varying sizes. We introduce simulation-based Benfordness estimation (SBBE), a Bayesian framework that models data as a mixture of Benford-conforming and non-conforming components. SBBE provides uncertainty-aware estimates, supports interpretability, and enables principled comparisons across datasets. We validate the approach using simulations and real-world Benford-compliant data and apply it to bioactivity datasets, where it revealed irregularities and sources of deviation. These results demonstrate the value of SBBE as a practical proxy for assessing data quality across scientific domains.

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INTRODUCTION

Data are the cornerstone of modern research, but their true value depends on their integrity. Even minor errors can have major consequences. The failure of the Mars Climate Orbiter, for instance, was traced back to a simple mismatch in measurement units.[1] Similarly, inconsistencies within or across databases can undermine the integrity of scientific findings.[2–5] In historical hurricane-tracking databases, such inconsistencies led to decades of undercounted storms, complicating assessment of climate models.[6]

Ensuring data quality requires vigilance at multiple levels, from the accuracy of individual samples[7] to the harmonization and validation of entire dataset collections.[3,4] While small numbers of data samples can be checked manually, such approaches are infeasible at scale.

Today, computational methods are widely employed to detect data irregularities that may reflect errors, rare events, or unexpected patterns.[8–10] Common approaches include data clustering to identify outliers,[8,11,12] comparison with historical data to reveal temporal shifts,[9] and isolation forests that flag datapoints easily separable from the bulk of the data.[13] Building on these foundations, more advanced techniques extend irregularity detection to the dataset level. For example, summary statistics or learned features from meta-learning enable clustering of datasets,[14] while active learning refines noisy labels by selecting ambiguous cases for expert review.[15] Yet, these approaches intensify the curse of dimensionality, as each dataset aggregates already high-dimensional samples, complicating both detection and interpretation.[16] This highlights a critical gap: the lack of scalable, domain-agnostic heuristics to flag irregularities at the dataset level.

An underexplored yet promising approach for uncovering hidden irregularities in datasets is Benford's law.[17,18] This mathematical principle posits that in naturally occurring datasets, smaller digits are more likely to appear as leading digits. For example, numbers beginning with "1" occur more frequently than those starting with "2", which in turn are more common than those starting with "3", and so forth.[17,18]

Data from a wide range of domains have been found to follow Benford's law, particularly when values span several orders of magnitude. Because this pattern appears so consistently, Benford's law has been employed to flag irregularities such as data noise, anomalies, or even potential fraud. In finance, it supports auditing by identifying suspicious patterns in transaction records.[19] In public health, it was used during the COVID-19 pandemic to assess the reliability of reported infection rates, revealing potential data manipulation.[20,21]

More recently, Benford's law has shown promise for flagging data generated by artificial intelligence (AI) systems. Several studies[22–24] employed Benford's law to distinguish AI-generated from human-written text. In the image domain, Zago et al.[25] reported that images generated by adversarial neural networks deviate from Benford's law relative to natural images. Beyond text and images, Bantange et al.[26] used Benford's law to analyze the distribution of musical note frequencies. They were able to distinguish human-composed music (e.g., by Johann Sebastian Bach) from AI-generated music that emulates Bach's style.

Importantly, deviations from Benford's law should not be viewed as evidence of data manipulation by default. Routine methodological factors can readily shift the observed first-digit frequencies away from the Benford distribution. These include rounding practices, binning strategies, and limited dynamic ranges, all of which are common in scientific measurement pipelines and can produce systematic distortions.[27] Yet, deviations can also be informative: in geophysics, the low signal-to-noise characteristics of marsquake recordings introduce departures from Benford's pattern that can be exploited for event detection.[28] Thus, interpreting Benford irregularities requires distinguishing artifacts of data processing from domain-specific signal features.

A variety of methods exist for quantifying a dataset's conformity with Benford's law ("Benfordness," see the "terminology" subsection of the methods section). These include classical goodness-of-fit (frequentist) tests, such as the chi-squared test,[29,30] and Bayesian model comparison measures, such as the Bayes factor (BF)[31] (supplemental methods). The chi-squared test compares the observed and expected (Benford) first significant digit (FSD) frequencies by calculating the squared differences between them. The BF, on the other hand, measures how strongly the data support a Benford model relative to an alternative model.

Although grounded in different statistical frameworks, both frequentist tests and Bayesian approaches exhibit similar limitations when used to assess conformity to Benford's law. Both are sensitive to sample size and tend to produce inflated results for large datasets, even with minor deviations.[30] Moreover, both approaches yield single-number outputs (chi-squared test and BF) that are not directly comparable across datasets of different sizes.

For the chi-squared test, one approach to rescale the test statistic is to use relative frequencies (proportions) rather than absolute frequencies (raw counts).[29,32,33] However, this violates the assumptions of the statistical test and can produce misleading results.[30,34]

Beyond generic goodness-of-fit methods, Leemis[35] formalized a Benford-specific statistic assessing adherence to Benford's law based on the most deviant FSD proportion. This made discrepancies easy to localize and explain (e.g., too many 5s as an FSD). Morrow[36] built on this by adding a sample-size-based scaling factor to improve small-sample reliability. However, these methods still yield a single value and may flag large datasets based on minor deviations.

To evaluate the different Benfordness measures, studies typically use Monte Carlo simulations.[37] Under a specified data-generating model, repeated samples are drawn, and the statistic of interest is computed. Because the generating model is known, the strengths and weaknesses of Benfordness measures can be quantified. Typically, a common setup considers mixtures of Benford and uniform FSD distributions, where the uniform component proportion is interpreted as contamination.[37,38]

Similarly, in practice, data often originate from multiple sources, such as different laboratories or different databases. Consider a dataset composed of two underlying sources: one that follows Benford's law and one that does not. If it were known how much of a dataset comes from a source following Benford's law and how much comes from a source that does not, the Benfordness of a dataset (i.e., one minus the contamination or, in other words, the data sample's proportion coming from Benford's FSD distribution) could be quantified directly.

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Viewing Benfordness through the lens of mixed-source contributions enables the calculation of a dataset's equivalent contamination proportion (ECP) (supplemental methods). The ECP is the proportion of contaminated observations such that the expectation (under this contaminated distribution) of a Benfordness measure equals the observed Benfordness measure. It has a closed-form solution for additive Benfordness measures under a fixed calibration. While the ECP addresses issues with the excess power of earlier methods,[39] it still reports a point estimate and lacks built-in uncertainty quantification. Moreover, the mixture perspective opens an underused avenue: the opportunity to incorporate domain knowledge. In practice, such contextual information is available, yet it is rarely encoded explicitly in Benfordness assessment.

Contextual information can not only improve the assessment of Benfordness but also be used to evaluate the quality of a Benfordness assessment method by determining whether its outcomes conform to known data quality criteria. For example, if a subset of data is identified as irregular through orthogonal investigations, removing it should increase Benfordness.

A domain that follows Benford's law and allows filtering based on contextual information is bioactivity data (i.e., measurements of the strength of interaction between small molecules and their biological targets). Bioactivity values range from picomolar to micromolar and are typically measured in terms of quantities exhibiting a log relationship to the Gibbs free energy of binding (ΔG). Given that ΔG is uniformly distributed, the measurement follows a log-uniform distribution, which theoretically follows Benford's law.[40] Furthermore, empirical evidence shows that bioactivity data adhere to Benford's law.[32,33] At the same time, bioactivity data are rich in metadata, allowing filtering depending on known data quality criteria (e.g., measurement type, stringency of data curation, and annotations in databases) to assess Benfordness.

In this work, we offer a Bayesian perspective on Benfordness assessment through mixed-source contributions and introduce the simulation-based Benfordness estimation (SBBE) approach for dataset-level anomaly detection. SBBE provides a probabilistic estimate of the Benfordness and enables the introduction of priors, making the assessment more stringent or relaxed depending on the stakes of the application. In addition to simulations, we evaluate SBBE using bioactivity data, demonstrating that it ranks data subsets according to established quality criteria. Moving to the assay dataset level, we identify irregularities even within maximally curated data and trace them to specific mechanisms ranging from benign reporting conventions to curation errors, highlighting SBBE's ability to guide expert attention.

RESULTS

The results section is organized into three parts. First, we introduce and validate the SBBE method (Figure 1), which addresses the limitations of standard statistical tests. Second, we show that SBBE predictions align with established data-quality criteria both within and across the major bioactivity databases. Finally, at the assay dataset level, we demonstrate how SBBE can guide experts in identifying sources of data irregularities related to conventional data reporting and curation errors.

SBBE enables Benfordness comparison across datasets of different sizes

To illustrate how dataset size affects Benfordness assessment, we simulated datasets that vary (1) in size and (2) in the proportion of FSDs drawn from Benford's distribution versus a uniform distribution. For each dataset, we computed the BF (i.e., the likelihood ratio of the data under two competing models) as a Benfordness measure to determine whether the data more closely align with Benford's distribution or a general multinomial distribution, representing any alternative FSD distribution.

For datasets with sample sizes below 1,000, the BF acted as expected: it tended to favor the Benford model when more than half of the data originated from Benford's distribution (dark green pixels in Figure 2A). In other words, the BF correctly favored the Benford model when most data points followed Benford's law. However, once the sample size exceeded 1,000, the BF required a considerably higher share of Benford-distributed data (up to 98%) to favor the Benford model (Figure 2A). At this scale, the BF consistently favored the more flexible reference distribution, even though most of the data were generated by the Benford model. Better calibration methods are, therefore, necessary to enable meaningful comparisons across datasets of different sizes.

To address the dependency of Benfordness estimation on dataset size, we used simulations. Specifically, we computed the BF across synthetic datasets of varying sizes and mixing ratios between Benford and uniform FSD distributions, which we refer to as given Benfordness. In other words, a given Benfordness value of 0.7 corresponds to a simulated dataset with 70% of its FSDs coming from a Benford distribution and 30% coming from a uniform distribution. We generated synthetic datasets with increments of 0.02 in given Benfordness.

For each sample size and given Benfordness, we generated 1,000 synthetic datasets and computed the BF (Figure 1C), producing a distribution for each setting (Figure 1B). To infer the given Benfordness for a real dataset, we compared its observed Benfordness measure with the distribution expected for its size, taking prior information into account (Figure 1A). We call this Bayesian method for Benfordness assessment SBBE.

To build intuition for SBBE, we present examples illustrating how calibration stabilizes Benfordness estimates across sample sizes (Figure 2B; note that for visualization purposes, we display a line connecting the histogram values). Specifically, we simulated five datasets with a fixed given Benfordness of 0.7, while varying only the number of observations. For each dataset, our framework yielded a posterior probability distribution over the given Benfordness, and we define the estimated Benfordness as its maximum *a posteriori* value (peak). As shown in Figure 2B, increasing the sample size made this probability distribution more concentrated around 0.7, showing no systematic bias as the sample size increased. Instead, SBBE reliably recovered the given Benfordness of 0.7 across all dataset sizes. This demonstrates that our simulation-based framework successfully accounts for dataset size, enabling fair and interpretable comparisons across datasets of varying sizes.

While SBBE can calibrate various statistics (e.g., chi-squared, Kuiper, etc.), we calibrated the BF in the simulated example (Figure 2B). BF was selected as the Benfordness measure

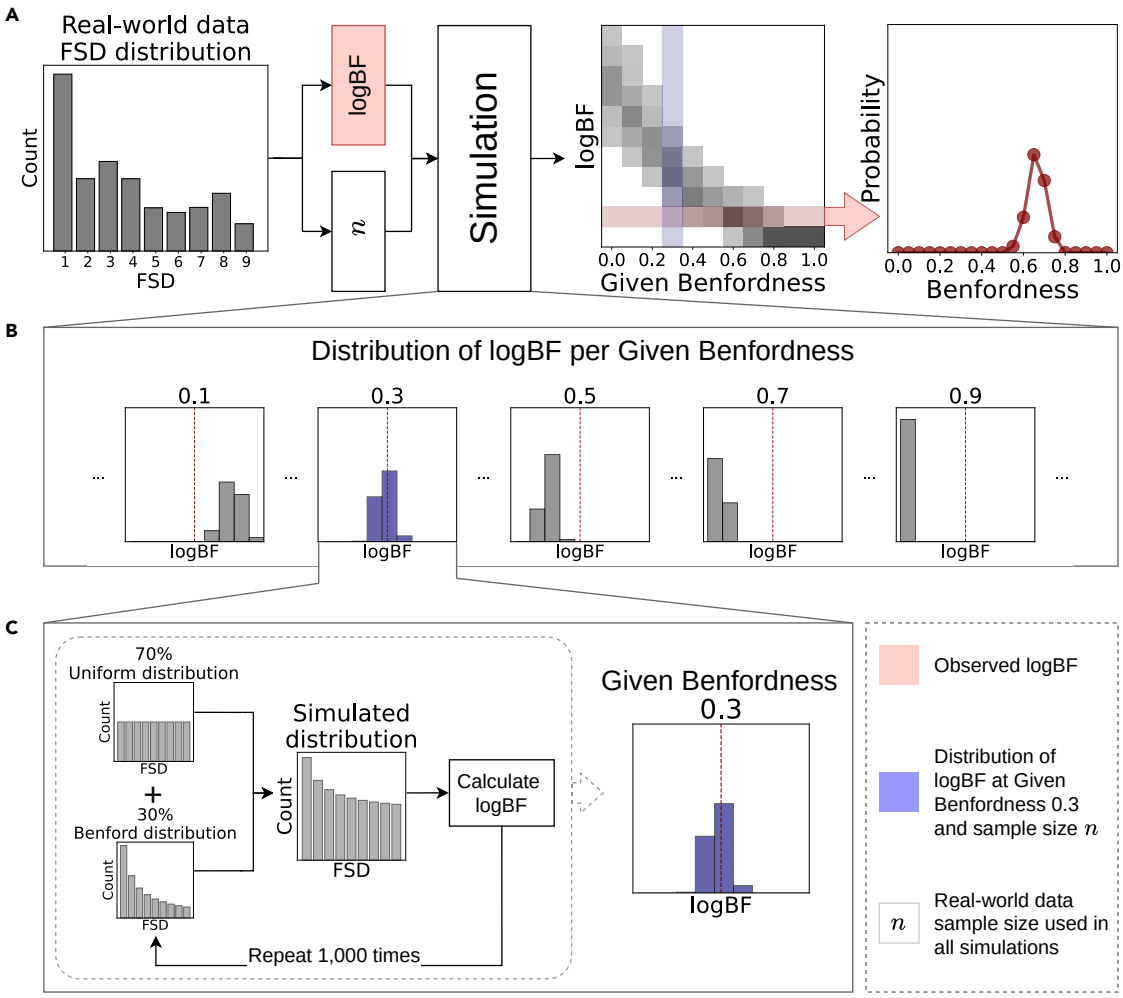


Figure 1. SBBE workflow

Simulation-based Benfordness estimation (SBBE) workflow.

(A) SBBE requires only two inputs from a real-world dataset: the sample size *n* and the observed logBF. The simulation produces a distribution of logBF values for each simulated given Benfordness, i.e., the true simulated proportion of Benford data (see section "terminology"). The blue band in the heatmap shows the logBF distribution at a given Benfordness of 0.3. We infer Benfordness by comparing the observed logBF to simulations (red band in the heatmap), yielding a probability density over Benfordness. The points in the density plot represent the actual computed values. The underlying data are a discrete probability distribution, represented as a curve for easier understanding. Estimated Benfordness (see section "terminology") is the maximum *a posteriori* value (peak in the Benfordness probability distribution plot).

(B) Simulated distributions of logBF for different given Benfordness values.

(C) Illustration of acquiring the distribution of logBF values for a given Benfordness value. Specifically, logBF is calculated for 1,000 simulated datasets with sample size *n* and given Benfordness 0.3.

because it recovers the given Benfordness well while maintaining a low overestimation probability. Overestimation of Benfordness can be harmful for irregularity detection as it may cause genuinely low-Benfordness datasets to appear benign. Comprehensive simulation-based calibration experiments were performed to compare various statistics (see Note S1 and Figure S1).

We also validate our approach on four example datasets known to follow Benford's law from various scientific domains (computer science, sociology, astronomy, and geography), illustrating the importance of uncertainty estimates for assessing Benfordness in the Note S2 and Figure S1.

Benfordness of bioactivity data aligns with data quality

We have shown that SBBE enables comparisons between datasets of different sizes. Here, we use bioactivity data to demonstrate that SBBE predictions align with established data quality criteria both within and across the major bioactivity databases, PDBbind,[41] ChEMBL,[42] and PubChem.[43]

We found that PDBbind exhibited the highest overall estimated Benfordness (96%), followed by ChEMBL (76%) and PubChem (42%) (Figures 3A and 3B). This trend aligns with the distinct data deposition concepts and curation standards employed in the compilation of these databases: PDBbind is a highly curated database of target-ligand bioactivity data, in which all data are

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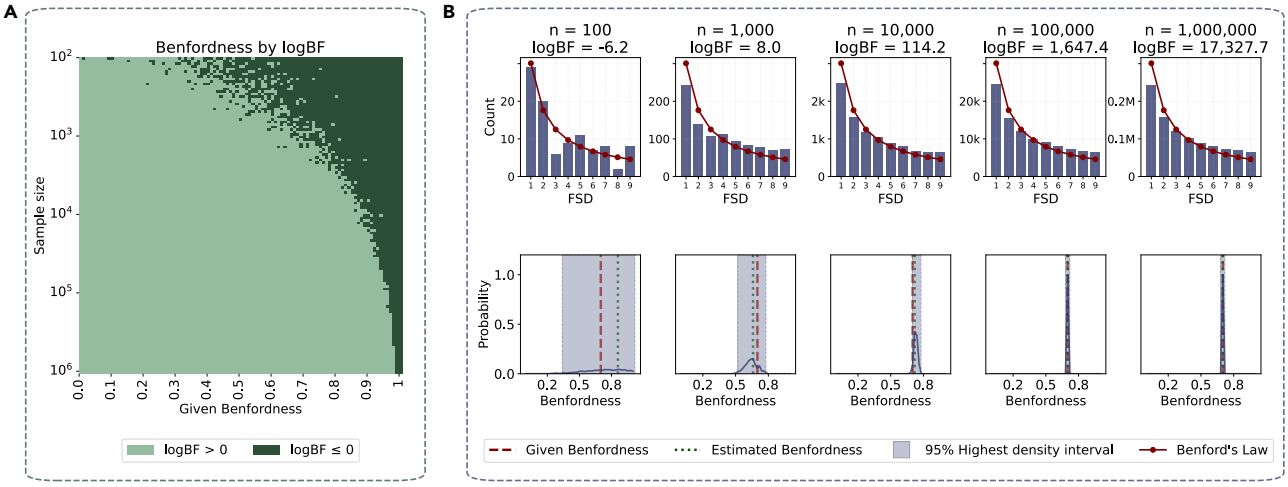


Figure 2. Size dependency and simulations

Size dependency of the Bayes factor (BF) in Benfordness testing and how it is addressed by SBBE.

(A) BFs for simulated samples of varying size, drawn from mixtures of a Benford and a uniform distribution over the first significant digits (FSDs). The plot shows that larger datasets require a higher proportion of Benford-conforming points to favor the Benford model (dark green) over the flexible reference model (light green).

(B) Illustration of the effect of calibration on assessing Benfordness using simulated datasets with a fixed given Benfordness of 0.7 and varying sample sizes. As the sample size increases, the estimated Benfordness becomes more sharply concentrated around the true value (red dashed line), demonstrating that the simulation-based framework effectively removes size-dependent bias and supports accurate inference. The blue-shaded area represents the highest density interval (95% HDI).

manually extracted from the literature (26,983 datapoints).[41] ChEMBL is a large bioactivity data resource that employs an expert-assisted data curation framework. As such, it represents a balance between high curation standards and data heterogeneity. PubChem operates as an open-access repository that accepts data submissions from the scientific community.

To investigate whether SBBE aligns with data quality within a database, we focused on ChEMBL for its rich annotations and balance between data volume and curation quality. We partitioned the ChEMBL data into two metadata categories expected to differ in data quality (i.e., measurement type and data source) and calculated the estimated Benfordness for each subset.

We stratified ChEMBL bioactivity data by bioactivity measurement type, focusing on six endpoints relevant to drug discovery. IC_{50}, EC_{50}, K_i, and K_d are determined in dedicated dose-response or binding assays. AC_{50} and potency, in contrast, are half-maximal concentrations fitted to high-throughput screening curves, often extrapolated beyond the tested concentration range.[44] The four directly measured endpoints showed high estimated Benfordness (>80%), whereas AC_{50} (66%) and potency (18%) showed substantially lower values (Figure 3C). Low estimated Benfordness, therefore, concentrates at endpoints inferred from screening curves rather than measured directly.

ChEMBL curates bioactivity data from multiple sources. The peer-reviewed literature is the largest, with approximately 2.5 million entries from over 300,000 assay datasets. PubChem is the second largest, contributing over 1.5 million entries. This subset, however, originates from only 1,647 assay datasets provided by 12 depositors and consists largely of dose-response endpoints from high-throughput confirmatory screens.

Literature-derived records in ChEMBL showed high estimated Benfordness (96%), whereas PubChem-derived records showed 28%. Consistent with the measurement-type analysis, low estimated Benfordness is concentrated in screening-derived data. Estimated Benfordness for all ChEMBL data sources is shown in Figures S5 and S6.

Additionally, we partitioned the ChEMBL data by publication year. Our analysis yielded consistently high estimated Benfordness across years (Figures S7 and S8), in contrast to the quality-related categories above.

We also assessed whether the presence of congeneric ligands (i.e., a series of structurally related molecules, commonly observed in bioactivity datasets) can have a substantial effect on the estimated Benfordness. However, excluding congeneric ligands from ChEMBL only had a negligible impact on estimated Benfordness (Note S3 and Figure S2).

SBBE enables finding irregularities in the data

Low estimated Benfordness values identify data anomalies but do not explain their existence. In this section, we show how SBBE can be used to prioritize expert inspection, thereby aiding the identification of irregularities in data. To this end, we shift the focus of our analysis from patterns to causes, tracing specific mechanisms that account for the observed deviations. For ChEMBL bioactivity data, this meant shifting our analyses from the global database level to the assay dataset level.

To ensure that Benfordness testing is applicable, we restricted our analyses to assay datasets with sufficient sample size (more than 80 data points[36]) and dynamic range (bioactivity values spanning at least two orders of magnitude[40]). Only 1% of the assay datasets in ChEMBL (4,308 of 350,417) satisfy both criteria. Despite their small proportion, the included assay datasets cover 44% of all bioactivity measurements in ChEMBL, reflecting the fact that a small number of assay datasets contribute

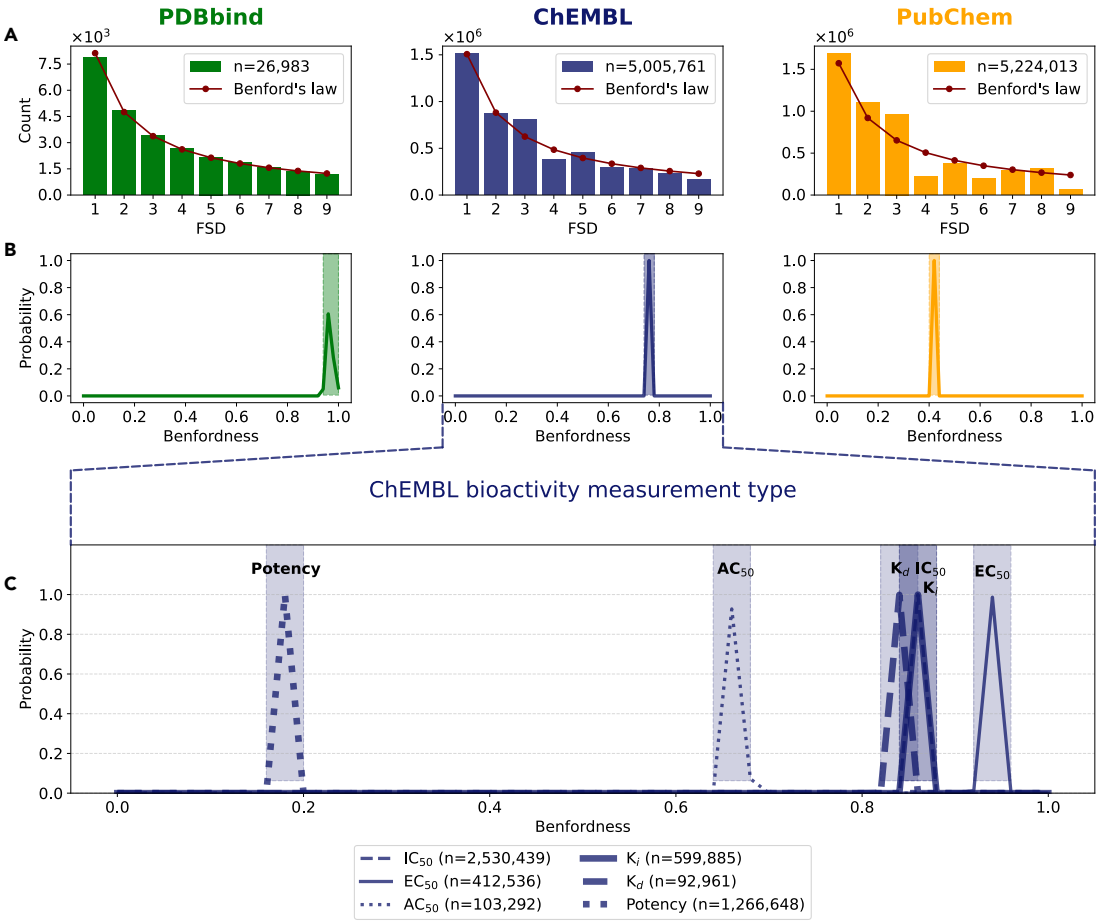


Figure 3. Benfordness of bioactivity databases

Analysis of the Benfordness of data from PDBbind, ChEMBL, and PubChem. The shaded areas in (B) and (C) represent the 95% HDI.
(A) Distributions of the first significant digits (FSDs) across the three databases, demonstrating different alignments with Benford's law.
(B) Estimated Benfordness by SBBE for the individual databases. Datasets compiled with higher curation standards align more closely with Benford's law.
(C) Across the bioactivity measurement types, potency data deviate the most from Benford's law (only approximately 18% estimated Benfordness). All other measurement types of bioactivity data align better with the expected distribution (EC_{50}: 94%; IC_{50}: 86%; K_i: 86%; K_d: 84%; AC_{50}: 66%).

a large share of the data. Moreover, the 4,308 selected assay datasets originate from 23 of the 34 data sources represented in the database, indicating that the filtered set remains diverse despite its reduced size.

To use SBBE for flagging assay datasets for priority inspection, we converted the probabilistic Benfordness estimate into a decision criterion by setting a threshold of 0.5. Specifically, for each assay, we summarized the SBBE posterior distribution by its median (i.e., the Benfordness value at which the cumulative posterior distribution reaches 50%; SBBE Benfordness; section "terminology"). We use SBBE Benfordness (calculated from the cumulative density) instead of estimated Benfordness (maximum *a posteriori*) because uncertainty tends to be higher at the assay dataset level due to a smaller sample size (median = 139 bioactivity values). To this end, assay datasets with a posterior median below 0.5 were flagged. This approach incorporates model uncertainty into the decision criterion, which is not directly possible with previous methods. With this setup, 866 of the 4,308 assay datasets (20%) were flagged due to low estimated Benfordness (Figure 4A).

For 578 of the 866 flagged assay datasets, we identified likely mechanisms contributing to low estimated Benfordness and assigned them to one of six mechanisms we identified as contributing to low Benfordness in bioactivity datasets (Figures 4B–4G). We refer to these as attributed assay datasets. Moreover, we provide median SBBE Benfordness, median IQR_{1-99}, and median SBBE per category in Table S1. We also provide these characteristics per assay in Data S1 along with plots of the FSD distributions per assay in Data S2.

Mechanisms 1 and 2: Data quantization on the log_{10} scale

Manual assessment of the FSD distribution revealed two distinct patterns emerging in the 866 flagged assay datasets. We found that 285 of them (33%) were rounded to one decimal place on the log_{10} scale (Mechanism 1; Figure 4B). Similarly, 110 of them (13%) were rounded to 0.05 increments (Mechanism 2; Figure 4C). Such quantization of measured bioactivity data is typical for low-sensitivity, large-scale assay screens, and it has

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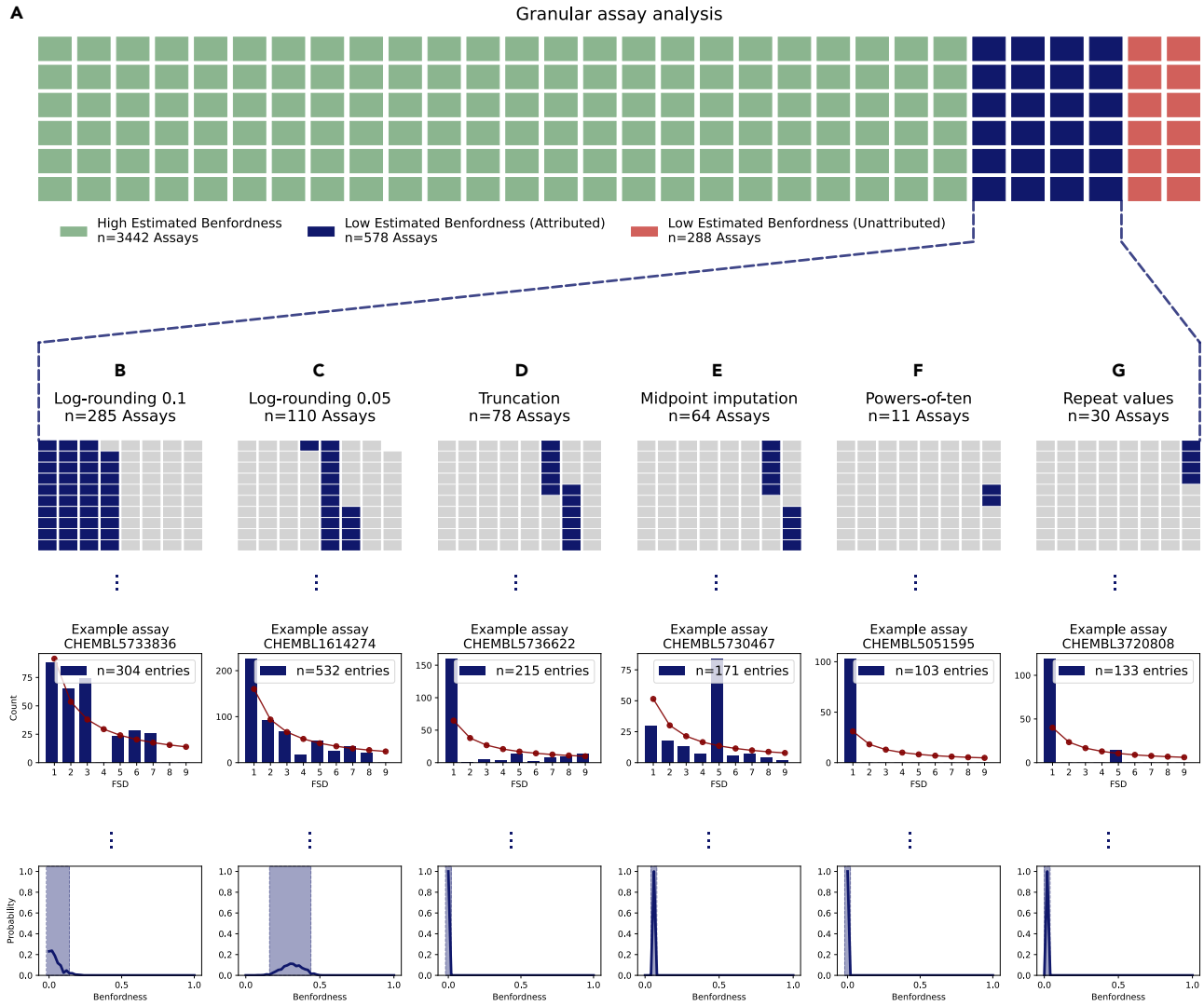


Figure 4. Granular analysis of ChEMBL assays

Assay-level analysis links low estimated Benfordness in ChEMBL to six mechanisms. The shaded areas in the bottom row represent the 95% HDI.

(A) Overview of the 4,308 assay datasets that meet the inclusion criteria (sample size; dynamic range) for Benfordness analysis. Green rectangles show assay datasets with high estimated Benfordness (n = 3,442), blue rectangles show assay datasets for which their low estimated Benfordness could readily be attributed to one or several mechanisms (n = 578), and red rectangles show assay datasets for which the reason underlying their low estimated Benfordness could not be readily identified (n = 288).

(B–G) Six mechanisms underlying low estimated Benfordness: (B) log-rounding to one decimal place, (C) log-rounding to 0.05 increments, (D) reporting censored values (lower or upper detection limit of assays) as exact values, (E) bioactivity value ranges reported as single midpoint values, (F) reporting ordinal bioactivity bins as boundary values, producing spikes at powers of ten, and (G) repeating specific bioactivity values disproportionately. The top row of diagrams visualizes the number of assay datasets assigned (by manual inspection) to a specific mechanism. The middle row overlays the assay datasets' first significant digit distributions with Benford's expectation, and the bottom row shows the corresponding estimated Benfordness using SBBE.

been shown that rounding or binning can distort the FSD distribution.[27]

The two quantization patterns are concentrated in specific data sources. In PubChem, 65% of entries show 0.05-step rounding and 15% show 0.1-step rounding. Together, 80% of the PubChem data included in the ChEMBL database are quantized. The low estimated Benfordness in PubChem data within ChEMBL was largely driven by assays using data quantization (1,647 assay datasets, 1,501,373 bioactivity values, estimated Benfordness = 28%), whereas PubChem assays without quantization showed higher values (1,322 assay datasets, 320,456 bioactivity values, estimated Benfordness = 76%).

Similarly, the data from the source GlaxoSmithKline had an estimated Benfordness below 2% because its entries fall entirely into the 0.1-step rounding category (Figures S5 and S6). Such cases emphasize that decisions based on Benfordness analysis must be grounded in an understanding of how data are produced and recorded, rather than treating Benford's law as an isolated data quality measure.

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In the remaining four categories, one or more FSDs dominated the distributions, leading to low estimated Benfordness. Although the outcomes appear similar, we can identify distinct causes for each.

Mechanism 3: Truncation

Biological assays have detection limits. As a result, bioactivity values beyond detection limits are often annotated as "greater than X" or "smaller than X." If such values are mistakenly annotated as "exact X," they cause an accumulation of bioactivity values corresponding to these limits. Among the 866 flagged assay datasets, 78 (9%) showed a dense cluster at a single high or low bioactivity value (Figure 4D). The estimated Benfordness of these flagged assay datasets combined was 54%. Removing the dense clusters increased the estimated Benfordness to 74%, indicating that the anomaly is driven by mislabeled boundary values rather than the underlying measurements.

Mechanism 4: Midpoint imputation

Among the 866 flagged assay datasets, 64 (7%) showed pronounced digit-heaping, characterized by repeated 5-based values (e.g., 0.55, 55.0, 5,500 nM; Figure 4E). Tracing these assay datasets showed that they were imported from patents into BindingDB and subsequently imported from BindingDB into ChEMBL. In the source patents, some bioactivity values are reported as ranges (e.g., 100–1,000 nM), and BindingDB replaces each range with its midpoint value, creating an artificial excess of the leading digit 5.

Mechanism 5: Powers of ten

Eleven of the 866 flagged assay datasets (1%) displayed values concentrated at powers of 10 (e.g., 0.1, 100, 10,000 nM; Figure 4F). Tracing these assay datasets showed that they originate from patents reporting bioactivity values in ordinal bins corresponding to concentration ranges rather than exact values. When imported into ChEMBL, each range is represented by a single boundary value, creating artificial spikes at the frequency of the FSD "1."

Mechanism 6: Repeat values

We observed that in some assay datasets, a small set of exact values occurs disproportionately often, leading to low estimated Benfordness. This pattern appeared in 30 of 866 flagged assay datasets (3%), for several reasons. For instance, assay CHEMBL3720808 (Figure 4G) contains 133 bioactivity values but only three distinct IC_{50} values (0.01, 101, and 501 nM). These values appeared 60, 59, and 14 times, respectively. The source publication[45] for assay CHEMBL964698, on the other hand, reports bioactivity values by grouping structurally related molecules, which leads to repeated values in ChEMBL. In other cases, the available metadata did not allow a definitive attribution of the reasons for value repetition.

Distinguishing genuine anomalies from data-limited assays

Of the 866 flagged assay datasets, 288 remained with no clear mechanism that can describe their deviation based on manual inspection, and we refer to these as unattributed assay datasets. Seventy percent of these unattributed assay datasets either have a below-median sample size or a narrow inter-quartile range of bioactivity values (IQR_{1-99} less than 2 on the $\log_{10}$ scale), compared to 35% among the attributed assay datasets and 51% among the non-flagged assay datasets. Low sample size and narrow dynamic range are known to limit the applicability of Benfordness assessment.[36,40] Whether a flagged assay represents a genuine anomaly or merely a data-limited case is a distinction that is not evident from a point estimate.

SBBE resolves the distinction between the data-limited case and the genuine anomaly by returning a posterior rather than a point estimate. Among the flagged assay datasets, those we could attribute to a specific mechanism often yielded sharply peaked posteriors, whereas unattributed assays often yielded high-uncertainty posteriors spanning the entire Benfordness range (see Figure S3B), which is a signature of weak evidence. A single-point measure assigns both cases comparably low scores and cannot distinguish them. While SBBE's single-point estimate aligns closely with the recently proposed ECP[39] (Figure S3A), SBBE's uncertainty separates genuine deviations from cases where the data are too limited to support a conclusion, directing expert effort away from uninformative datasets. Moreover, SBBE allows users to utilize prior information to tailor anomaly screening workflows to specific domains. We demonstrate the use of priors in SBBE in Figure S4 for both restrictive and permissive screening at the assay-dataset level.

DISCUSSION

In this work, we introduce SBBE, a Bayesian framework for assessing Benfordness. By simulating first-digit distributions from mixtures of Benford and non-Benford (uniform) distributions, SBBE provides an intuitive reference for how "Benford-like" a dataset is. Rather than producing a binary accept/reject outcome, SBBE estimates the degree of conformity, yielding an interpretable percentage of Benfordness that can be compared across datasets of different sample sizes. Importantly, SBBE returns a posterior distribution over Benfordness, providing an uncertainty measure that allows researchers to distinguish confident deviations from cases with weak evidence. Finally, the Bayesian formulation enables the incorporation of prior assumptions about the expected Benford behavior, allowing the assessment to be more restrictive or permissive, depending on the screening objective and domain context.

Applied to major bioactivity databases, SBBE produced a Benfordness ranking consistent with differences in curation intensity, with PDBbind ranking highest, followed by ChEMBL, and PubChem lowest. Within ChEMBL, SBBE clearly distinguished well-behaved subsets of the database from those that deviate from Benford's law. A closer inspection of the deviating subsets identified six mechanisms: some reflected benign reporting conventions, while others reflected data-quality issues warranting further review.

Importantly, deviations from Benford's law are neither necessary nor sufficient evidence for poor data quality: legitimate measurement protocols or bounded variables can produce deviations, while erroneous or manipulated data may still conform to Benford's law. SBBE, therefore, provides an indicator of where to look, not an automatic verdict. Its applicability is also constrained by sample size and dynamic range, with only 1% of all ChEMBL assays meeting both criteria in our analysis.

Future work could extend SBBE beyond the settings explored here by calibrating alternative statistics. It could also incorporate

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domain knowledge through informative priors in both drug discovery and other application domains.

We hope SBBE will help the scientific community integrate lightweight, domain-agnostic integrity checks into routine data workflows, improving the interpretability and trustworthiness of shared resources. To facilitate adoption, we provide a well-documented, ready-to-use Python package implementation in our public repository.

METHODS

Datasets

ChEMBL36

ChEMBL36 was retrieved as an SQLite database. Bioactivity data were selected from the ACTIVITIES table when they met all the following criteria: (1) STANDARD_TYPE (bioactivity measurement type) was one of IC_{50}, EC_{50}, AC_{50}, K_i, K_d, or potency; (2) STANDARD_RELATION was equality (=), to exclude censored bioactivity values beyond the detection limit (e.g., ">" and "<"); and (3) STANDARD_UNITS were nanomolar (nM). The numerical value used for Benfordness analysis was STANDARD_VALUE (in nM). In addition to the retrieved bioactivity data, assay-, target-, document-, and source-level metadata was retrieved, including the data source (ACTIVITIES.SRC_ID) and publication year (DOCS.YEAR). The complete list of extracted metadata fields and the SQL query are provided in the project repository (github.com/molinfo-vienna/sbbe_paper/[46]). Overall, we retrieved 5,005,761 bioactivity values from 350,417 assay datasets.

PDBbind

PDBbind (v.2024) protein-ligand bioactivity data were obtained under a paid subscription, and data retrieval was restricted to the protein-ligand "general set." The general set is a large collection of 26,983 protein-ligand complexes and their associated bioactivity data.

PubChem

PubChem bioactivity data were retrieved via bulk download from the PubChem File Transfer Protocol (FTP) site hosted by the National Center for Biotechnology Information (NCBI: ftp.ncbi.nlm.nih.gov/pubchem/). Specifically, we used the Concise assay tables, parsed them into a unified tabular format, and retrieved bioactivity data for all measurement types to conduct a global Benfordness assessment. Overall, we retrieved 5,224,013 bioactivity values from 473,567 assay datasets.

Statistical analysis

We developed a software framework to quantify deviations from Benford's law in bioactivity data using simulation-based inference. Our approach simulates distributions for a given measure of Benfordness under mixtures of Benford and uniform FSD distributions and then estimates the most likely mixture ratio and its uncertainty using posterior inference (Figure 1).

Simulation of Benford-uniform mixtures

We simulated synthetic datasets by drawing samples from mixtures of Benford and uniform distributions over FSDs (1–9). The Benford component followed the theoretical distribution $P_b(d) = log_{10}(1 + 1/d)$, while the uniform component assigned equal probability to each digit. We selected the uniform distribution as a reference because previous studies suggest that uniformity over the digits can indicate irregularity.[37,47,48] Furthermore, the uniform distribution is the most average multinomial distribution. While SBBE can be extended to use any distribution as a reference, we use the uniform distribution throughout this work for the reasons outlined above. For each dataset of interest, we drew samples corresponding to the dataset's size, with mixing ratios λ ranging from 0 to 1 in increments of 0.02. For each mixing ratio-dataset size combination, we drew R = 1,000 datasets and calculated a measure of Benfordness t_i, obtaining a kernel density estimate $\hat{p}(s|\lambda)$ for each mixing ratio-dataset size combination where K is a Gaussian kernel with bandwidth h obtained from the default implementation in SciPy:[49]

$$\hat{p}(s|\lambda) = \frac{1}{Rh}\sum_{i=1}^{R} K\left(\frac{s - t_i}{h}\right). \quad \text{(Equation 1)}$$

Posterior estimation over mixture ratios

To estimate the likely mixture ratio of an observed dataset, we first compute its Benfordness measure. For each mixing ratio and dataset size, we smooth the simulated measure of Benfordness distributions using kernel density estimation to obtain continuous likelihood functions (Equation 1). Utilizing a non-informative prior ($p(\lambda)$ = Beta(1,1)), these likelihoods are then used to evaluate the probability of the observed statistic across mixing ratios, enabling computation of an approximate posterior distribution $\hat{p}(\lambda|s)$ over mixture ratios:

$$\hat{p}(\lambda|s) = \frac{\hat{p}(s|\lambda)p(\lambda)}{\int \hat{p}(s|\lambda)p(\lambda)d\lambda}. \quad \text{(Equation 2)}$$

Terminology

We refer to Benfordness as a dataset's (typically unknowable) property of adhering to Benford's law throughout the manuscript, and we use Benfordness to annotate the x axis when plotting posterior densities. We refer to given Benfordness as the mixing ratio used to simulate an FSD dataset. We refer to estimated Benfordness as the maximum *a posteriori* (λ where Equation 2 is maximal). We refer to SBBE Benfordness as the median of the posterior distribution.

RESOURCE AVAILABILITY

Lead contact

Requests for further information and resources should be directed to and will be fulfilled by the lead contact, Johannes Kirchmair (johannes.kirchmair@univie.ac.at).

Materials availability

Not applicable.

Data and code availability

- All data used in this work were sourced from ChEMBL36,[50] PubChem,[51] and PDBbind 2024.[52]
- The source code for SBBE is available as a Python package on GitHub at https://github.com/molinfo-vienna/sbbe_paper/, under the MIT license and archived on Zenodo.[46]

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AUTHOR CONTRIBUTIONS

U.A.-S., M.W., and J.K. conceived and designed the experiments, analyzed the data, contributed materials/analysis tools, and wrote the paper. U.A.-S. and M.W. performed the experiments.

DECLARATION OF INTERESTS

The authors declare no competing interests.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

In the course of preparing this work, the authors used ChatGPT 4 and 5, and Claude Sonnet 4, to improve the language. Subsequently, the authors carefully reviewed and edited the content as required, assuming full responsibility for the publication's content.

SUPPLEMENTAL INFORMATION

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Patterns, Volume 7

Supplemental information

A Bayesian framework for anomaly detection in scientific data based on Benford's law

Uday Abu-Shehab, Matthias Welsch, and Johannes Kirchmair

Supplemental Methods

Commonly used statistical measures of Benfordness

Benford's Law states that in many naturally occurring datasets, the leading digit is not uniformly distributed. Rather, probability of observing a leading digit d is $P_b(d) = \log_{10}(1 + 1/d)$.[S1,S2] These probabilities stem from a measure that is base and scale invariant.[S3] To assess how much a sample of size n conforms to Benford's law multiple measures have been proposed. In this section, we give an overview of previous works.

In the following, we denote c_i as the observed counts for digit i with a maximum of K in the empirical dataset $\mathcal{D}$. E.g., c_1 is equal to how often 1 occurs as a leading digit. Similarly, we define $p_i = c_i/n$ as the proportions of a digit i occurring in the sample. Furthermore, we define $F(k) = \sum_{i=1}^{k} p_i$ as the empirical distribution function, and $F_b(k) = \sum_{i=1}^{k} P_b(i)$ as the cumulative distribution function according to Benford's law.

Bayes Factor between Dirichlet-Multinomial and Multinomial with fixed probabilities according to Benford's law

Bayes factors assess the likelihood ratio given data D between two competing models H_0 and H_1. It has been suggested to compare data under a multinomial with fixed probabilities according to Benford's law (H_0) against a flexible Dirichlet-Multinomial (H_1).[S4] The Dirichlet-Multinomial model encodes prior beliefs about the multinomial probabilities via a Dirichlet distribution, with these prior estimates determined by the concentration parameters $\boldsymbol{\alpha}$. Each element α_i of the vector $\boldsymbol{\alpha}$ encodes prior information about category i, representing how often this entry was observed previously. As we have little prior knowledge, we set $\alpha_i = 1$ for all i. In our experiments, we calculate the logarithm of the Bayes factor S, which is numerically stable and has the following closed form solution:[S4]

$$S = \ln \frac{p(D|H_1)}{p(D|H_0)} = \overbrace{\sum_{i=1}^{K} \ln \Gamma(c_i + \alpha_i) - \sum_{i=1}^{K} \ln \Gamma(\alpha_i) - \ln \Gamma\big[\sum_{i=1}^{K} c_i + \alpha_i\big] + \ln \Gamma\big[\sum_{i=1}^{K} \alpha_i\big]}^{\text{Dirichlet-multinomial}} - \overbrace{\sum_{i=1}^{K} c_i \ln P_b(i)}^{\text{Multinomial with fixed probabilities}} .$$

χ^2 based on counts

The χ^2 statistic compares the observed counts of each leading digit to the expected counts under Benford's law:

$$\chi^2 = \sum_{i=1}^{K} \frac{(c_i - nP_b(i))^2}{nP_b(i)}.$$

Kolmogorov–Smirnov test

The Kolmogorov–Smirnov (KS) test evaluates how closely two distributions match. In the one-sample version, it compares an empirical cumulative distribution to a specified theoretical CDF

and uses the maximum vertical gap between them as the test statistic. To test whether a sample follows Benford's law, this becomes

$$D = \max_j |F(j) - F_b(j)|.$$

Kuiper's test

The Kuiper test, like the KS test, compares two distributions through their cumulative functions. Unlike KS, it accounts for both the largest positive and largest negative deviations. For testing conformity to Benford's law, it takes the form

$$V = |\max_j F(j) - F_b(j)| + |\min_j F(j) - F_b(j)|.$$

Leemis' maximum distance

Leemis' maximum distance[S5] focuses on discrepancies between the probability masses rather than the cumulative distributions. It measures the largest absolute difference between the empirical first digit frequencies p_j and the theoretical Benford probabilities $P_b(j)$:

$$m_l = \max_j |p_j - P_b(j)|.$$

Morrow's maximum distance

Morrow[S6] suggested scaling Leemis' maximum distance by $\sqrt{n}$ to ensure that the statistic converges in distribution to a continuous function of a normally distributed random variable with mean 0, facilitating inference. The resulting test statistic is

$$m_m = \sqrt{n} \max_j |p_j - P_b(j)|.$$

Cho-Gaines Euclidean distance

Unlike statistics that focus on the largest deviation, the Cho-Gains Euclidean distance[S7] incorporates information from all leading–digit probabilities:

$$d = \sqrt{n \sum_{j=1}^{K} (p_j - P_b(j))^2}.$$

Equivalent Contamination Proportion

Let T be a statistic measuring deviation from Benford's law, with $s = T(\mathcal{D})$ denoting the value of the statistic computed on the observed dataset $\mathcal{D}$ of size n. Define the contamination mixture distribution

$$\mathcal{M}(\phi) = \phi \cdot \mathcal{C} + (1 - \phi) \cdot \texttt{benford},$$

where $\mathcal{C}$ is a specified contamination distribution and $\phi \in [0, 1]$ denotes the contamination proportion. A sample of size n drawn from $\mathcal{M}(\phi)$ is denoted by $M \sim \mathcal{M}(\phi)$.

The Equivalent Contamination Proportion (ECP) is defined as the contamination proportion ϕ for which the expected value of the statistic computed on a sample from the mixture distribution equals the observed statistic:

$$\text{ECP} = \left\{\phi \in [0,1] : \mathbb{E}_{M\sim\mathcal{M}(\phi)}\left[T(M)\right] = s\right\}.$$

In other words, the ECP quantifies the proportion of contaminated observations in a hypothetical sample whose expected deviation from Benford's law matches the deviation observed in the empirical dataset, when both are measured using the same statistic T.[S8]

Computation of ECP for non-additive statistics

To compare SBBE and ECP in fair terms, we computed both from the same measure (Bayes factor). Because this statistic is non-additive, no closed-form expression for its expectation under a given contamination proportion is available. Therefore, we estimated the ECP using the Bayes factor, following the author's recommendations.[S8] Specifically, for each assay dataset $\mathcal{D}$ of size n with observed statistic $T(\mathcal{D})$, we first evaluated the statistic at the two extremes, $\phi = 0$ (fully non-conforming) and $\phi = 1$ (fully Benford-conforming). If $T(\mathcal{D})$ fell outside the interval spanned by these values, the ECP was set to the corresponding bound.

Otherwise, we searched for the contamination proportion ϕ whose expected statistic matched $T(\mathcal{D})$. Specifically, at each candidate ϕ, we simulated 1000 datasets of sample size n with that contamination proportion ϕ, computed the mean statistic across the simulations and increased ϕ in increments of 0.02 until we could no longer reject the null hypothesis that the mean equals $T(\mathcal{D})$ (one-sample t-test, significance level of 0.05). The resulting ϕ was the estimated ECP.

Supplemental Notes

Note S1: Calibration enables robust Benfordness assessment.
Note S2: SBBE enables robust Benfordness assessment on real-world data.
Note S3: Analogue bias does not change Estimated Benfordness meaningfully.

Note S1 Calibration enables robust Benfordness assessment

Having demonstrated that calibration of the BF enables meaningful comparisons across datasets of different sizes, we proceed to a more detailed evaluation of its capacity to recover the Given Benfordness in simulated datasets. Specifically, the same calibration framework was applied to seven commonly used statistics to compare how their calibrated estimates behave, and to select the best inference strategy for further evaluation. These statistics are: BF, Kuiper, Euclidean, Morrow, Kolmogorov-Smirnov, Chi-Squared, and Leemis (Figure S1a). Each of these statistics is explained in the Supplemental Methods.

We evaluated SBBE in two simulation settings: uniform and multinomial. In the uniform setting, test datasets were generated from Benford-uniform mixtures of FSD distributions, matching the distributional assumptions used for calibration. This assesses the accuracy of our method under controlled conditions. In the multinomial setting, test datasets were generated from mixtures of Benford and general multinomial distributions of FSDs, while calibration is performed using Benford-uniform mixtures. Because the test data deviated from the calibration model in the second setting, it provides a measure of robustness.

Across the uniform and multinomial simulation settings, we quantified performance using two complementary criteria: mean absolute error (MAE) and overestimation risk. MAE measures the absolute difference between the Estimated Benfordness and the Given Benfordness, treating underestimation and overestimation symmetrically. On the other hand, overestimation risk is defined as the proportion of simulation runs in which the Estimated Benfordness exceeds the Given Benfordness. This is particularly important because the most critical failure mode is overestimating Benfordness, i.e., incorrectly inferring that a dataset adheres to Benford's law when it does not.

Each simulation run was defined by three choices: simulation setting (uniform or multinomial), sample size, and Given Benfordness. For each such specification, we generated 1,000 datasets and evaluated MAE and overestimation risk using each inference strategy.

In the uniform setting, MAE decreased with increasing sample size across all strategies (Figure S1a). Specifically, errors dropped consistently to an MAE below 0.02 for datasets of size 5,000, i.e., smaller than the spacing between Given Benfordness increments. The overestimation risk also declined with increasing sample size and fell below 0.1 across all strategies (Figure S1a). Together, these results show that calibration yields accurate and conservative estimates when the data follows the calibration assumptions and when the sample size is large.

In the multinomial setting, performance was worse than in the uniform setting. This is expected because calibration assumes a Benford-uniform mixture, whereas the test data follows a more general multinomial mixture process. Even so, the MAE stayed below 0.15 for all strategies across sample sizes, indicating that calibration remains informative under model mismatch (Figure S1a). Unlike in the uniform setting, differences between strategies became more visible at larger sample sizes in the multinomial setting. For instance, the inference strategy based on Morrow's statistic achieved the lowest average MAE, but it also showed consistently high overestimation risk, remaining near 0.4 in this scenario (Figure S1a). By contrast, the BF- and Chi-Squared-based inference strategies exhibited lower overestimation risk, dropping below 0.1 at large sample sizes, while maintaining competitive MAE.

Building on the results of the MAE and overestimation risk assessment, we found a clear trade-off between inference strategies. The inference strategy based on Morrow's statistic achieved low MAE values but carries a high risk of overestimation. In contrast, the inference strategy based on BF combined low overestimation risk with moderately low MAE (around 0.15).

To investigate the trade-off between MAE and overestimation risk for inference strategies based on Morrow's statistic and BF, we plot the Given Benfordness against the Estimated Benfordness for simulated datasets of size 10,000 (Figure S1b). At high Given Benfordness, both

inference strategies based on Morrow's statistic and BF produced accurate estimates. In contrast, most discrepancies arose at low to intermediate Given Benfordness. The errors of the inference strategy based on Morrow's statistic were symmetric between over- and underestimation of Benfordness, whereas errors of the inference strategy based on BF were mostly underestimates (Figure S1b). For irregularity detection, overestimation can be harmful because it can make genuinely low-Benfordness datasets appear benign and reduce sensitivity to anomalies. Accordingly, we use the BF-based inference strategy in all subsequent analyses.

Moreover, we compared the performance of SBBE with ECP,[S8] as both methods frame Benfordness assessment within a mixture-based framework. Overall, SBBE and ECP[S8] show very similar performance and yield consistent conclusions (Figure S1a). However, SBBE provides additional insight by returning a Highest Density Interval (HDI), enabling more informed decisions based on the estimated uncertainty (Figure S3). In addition, as a Bayesian framework, SBBE allows the incorporation of prior assumptions, so users can adopt more restrictive or permissive Benfordness screening based on the application context. While all SBBE estimates in this manuscript are obtained with a non-informative prior, a visualization and further discussion of incorporating priors are provided in Figure S4.

Note S2 Simulation-based inference enables robust Benfordness assessment on real-world data

Having established that SBBE performs reliably on synthetic data (Section Note S1), we next applied it to real-world datasets. We drew on a publicly available dataset collection commonly used to illustrate Benford's law (`https://testingbenfordslaw.com/`). This collection aggregates example datasets from different domains, providing a convenient set of cases with varying degrees of Benfordness. We selected four representative cases to demonstrate the range of behaviors our method could capture: two that clearly followed Benford's law, one with borderline adherence, and one that is characterized by high uncertainty (Figure S1c).

We began with file sizes from the Linux 2.6.39.2 source tree. This dataset comprises 36,702 entries spanning five orders of magnitude, a characteristic that typically favors conformity to Benford's law. As expected, SBBE detected high Estimated Benfordness (98%) with minimal uncertainty. The second dataset represents a collection of world population samples collected over time. This dataset has 12,809 records and similarly exhibited a high Estimated Benfordness (86%) with minimal uncertainty.

Next, we examined the distances of stars from Earth, measured in light-years, a classic example often cited as naturally Benford-like. This dataset exhibited high Estimated Benfordness (98%), but also a higher degree of uncertainty. This outcome is intuitive: although the data spans several orders of magnitude (4 to 3000 light-years), the number of entries is relatively small (300 distances), allowing for statistical variation.

Finally, we examined the distances between major European cities in kilometers. Unlike previous examples, this dataset is small (comprising 210 distances between cities), and the distances are confined to a limited range (158 to 4,532 km). The deviation from the expected digit distribution led our method to identify only low Estimated Benfordness (54%), with considerable uncertainty.

Together, these examples illustrate how SBBE captures both the presence and strength of Benford-like structure, while transparently expressing confidence in its conclusions.

Note S3 Analogue bias does not change Estimated Benfordness meaningfully

Analogue bias is the tendency for multiple structurally similar compounds to be tested in the same assay. It arises from the greater synthetic accessibility of chemical series compared with randomly selected compounds. To ensure our analysis is not confounded by this bias, we performed a control experiment using ChEMBL.

First, we clustered all compounds in ChEMBL by their structure (BitBIRCH[S9,S10] using radius 2 Morgan Fingerprints with 2048 features and a threshold of 0.4). Given that 200,036 out of 365,313 clusters contained only one molecule, we analyzed the data stratified by minimum cluster size. Specifically, for each threshold $t \in \{1, 2, 5, 10, 20, 50, 100\}$, we included only clusters with size $\geq t$.

By sampling one activity value per included cluster, we reduced the influence of analogue bias because chemical series are typically contained within clusters. We also drew a sample of the same size (increasing the threshold t decreases the sample size) randomly from all activity values. We calculated the Estimated Benfordness for the "cluster" and "random" samples. We repeated this process 100 times and compared it to the overall Estimated Benfordness of ChEMBL (Figure S2). Since the maximum deviation in mean Estimated Benfordness across all cluster-size thresholds from the overall value was only 0.06, we conclude that analogue bias has a negligible effect on Estimated Benfordness.

Supplemental Figures

Figure S1: SBBE enables Benfordness comparison between datasets of different sizes.
Figure S2: Molecules analogue bias analysis.
Figure S3: SBBE and ECP agree as point estimates, but differ in what they report.
Figure S4: Incorporating prior knowledge into Benfordness assessment.
Figure S5: Adherence to Benford's law in ChEMBL partitioned by data source (FSD).
Figure S6: Adherence to Benford's law in ChEMBL partitioned by data source (SBBE).
Figure S7: Adherence to Benford's law in ChEMBL partitioned by year (FSD).
Figure S8: Adherence to Benford's law in ChEMBL partitioned by year (SBBE).

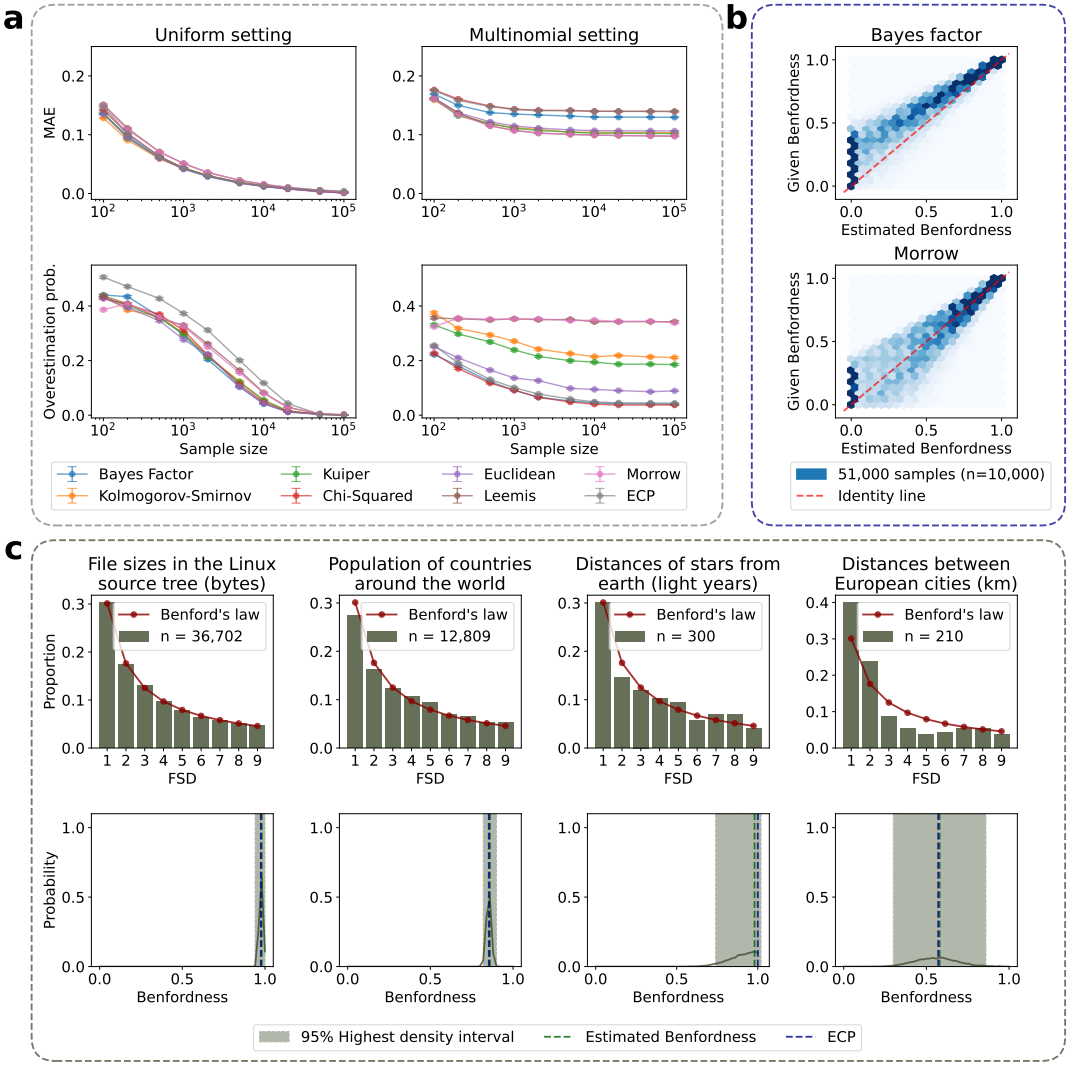


Figure S1: SBBE enables Benfordness comparison between datasets of different sizes. (a) Performance of our simulation-based calibration method across simulated datasets using seven Benfordness measures under two conditions: mixtures of uniform and Benford distributions, and mixtures of multinomial and Benford distributions. In addition to the seven calibrated Benfordness measures, we include the Equivalent Contamination Proportion (ECP)[S8] based on Chi-Squared, where Benfordness is 1 - contamination. Prediction accuracy, measured by the mean absolute difference (MAE) between the Estimated Benfordness and true mixture proportions (error bars calculated using normal approximation), and overestimation probability (Clopper-Pearson error bars) improved with sample size and remained robust even when calibration assumptions were violated. (b) Example of two inference strategies based on BF and Morrow's statistic: Density of Estimated Benfordness versus Given Benfordness for 51,000 simulated datasets (n = 10,000 samples). The identity line (red dashed line) represents perfect agreement between the Given Benfordness and the Estimated Benfordness. This figure shows clearly that BF tends to underestimate the Given Benfordness, while Morrow tends to symmetrically over- or underestimate it. (c) Validation of SBBE on real-world datasets (see Note S2). For each dataset, the empirical FSD distribution is shown (top), together with the Estimated Benfordness (bottom). The green dashed line indicates the Estimated Benfordness, and the light green background area indicates the highest density interval (95% HDI). Panel c clearly shows high agreement between SBBE and ECP, with SBBE providing Bayesian uncertainty estimation for more informed decision-making.

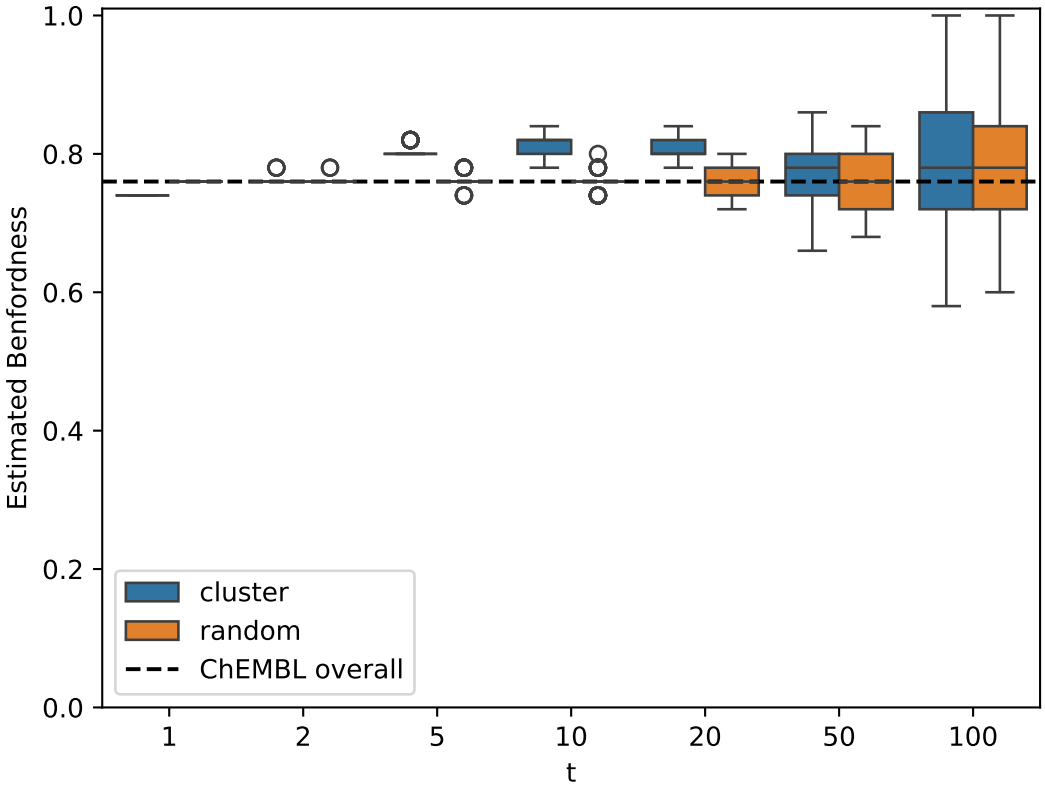


Figure S2: Estimated Benfordness values obtained from cluster-based and random sampling of ChEMBL activity values across different minimum cluster-size thresholds.

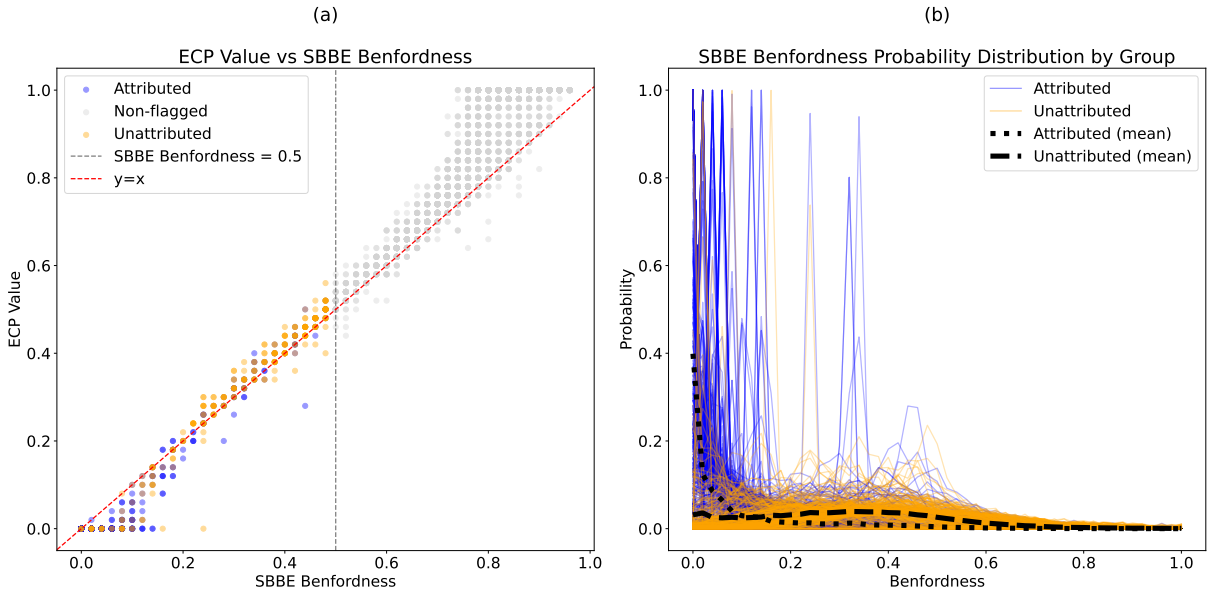


Figure S3: Comparison of SBBE and ECP across ChEMBL assay datasets. (a) SBBE point estimates of SBBE agree closely with ECP (dashed line, y=x). The computation of ECP is explained in Supplemental Methods. (b) Posterior Benfordness distributions per assay: assays attributable to a mechanism (blue) yield sharply peaked posteriors, whereas unattributed assays (orange) yield high uncertainty posteriors; bold black lines show group means.

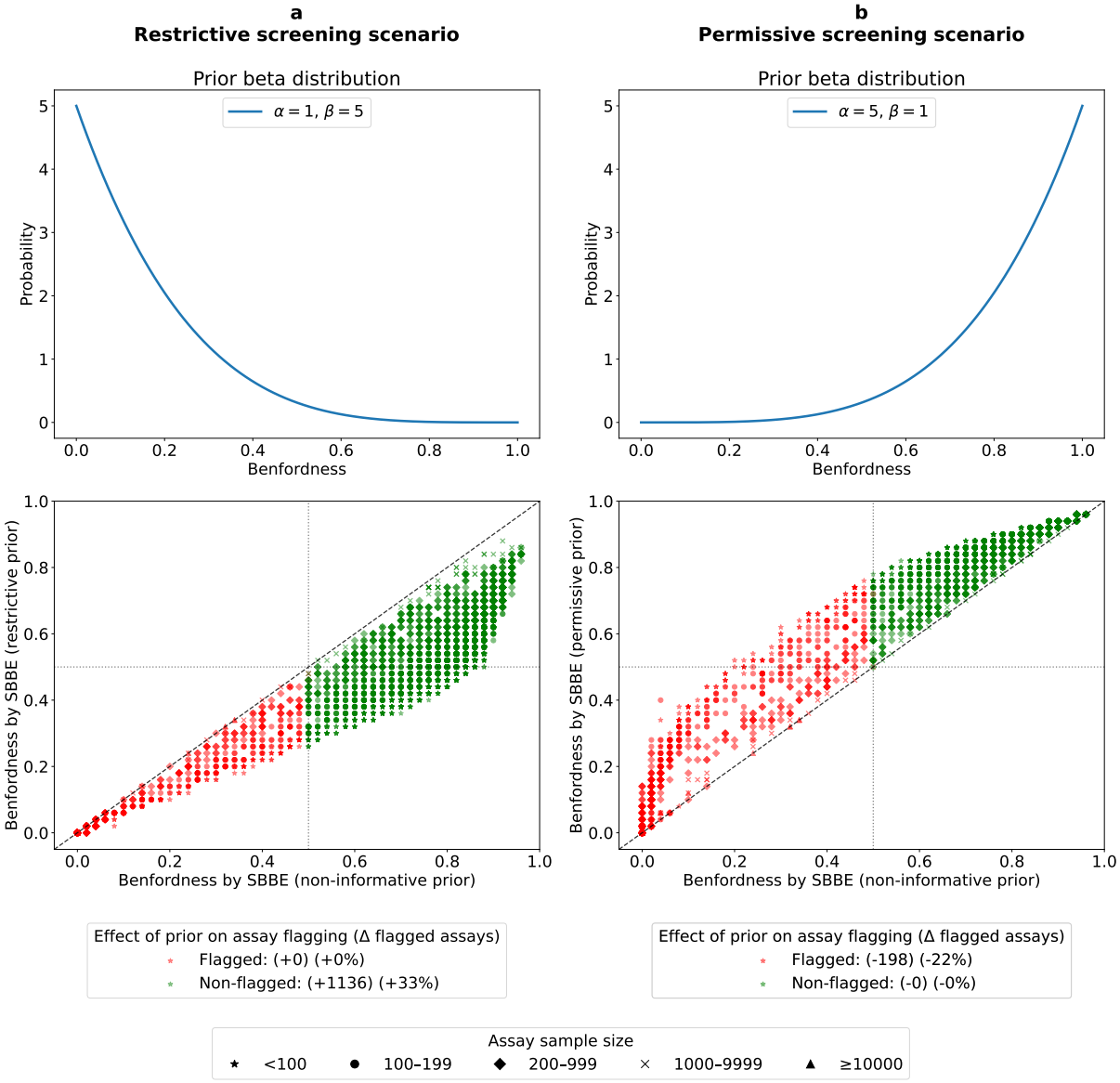


Figure S4: Benfordness assessment ChEMBL assays of using SBBE while incorporating prior knowledge. (a) Restrictive screening scenario using a Beta prior on Benfordness (α=1, β=5); (b) Permissive screening scenario (α=5, β=1). In each scenario, the top panel shows the corresponding prior Beta distribution, and the bottom panel compares assay-level Benfordness estimates obtained with the prior (y-axis) against the non-informative prior baseline (x-axis) across assays. The dashed diagonal indicates $y = x$, and the dotted lines mark the decision boundary (0.5). Point color indicates flagging decision by the baseline (non-informative prior) model, i.e., whether assays are non-flagged (green) or flagged (red). The legend reports the net change in flagged assay counts relative to the non-informative prior run (prior − non-informative prior). Marker shape denotes assay sample-size bins. This plot shows that assays with smaller sample sizes (e.g., stars with $<$100 entries) are most affected by the use of priors.

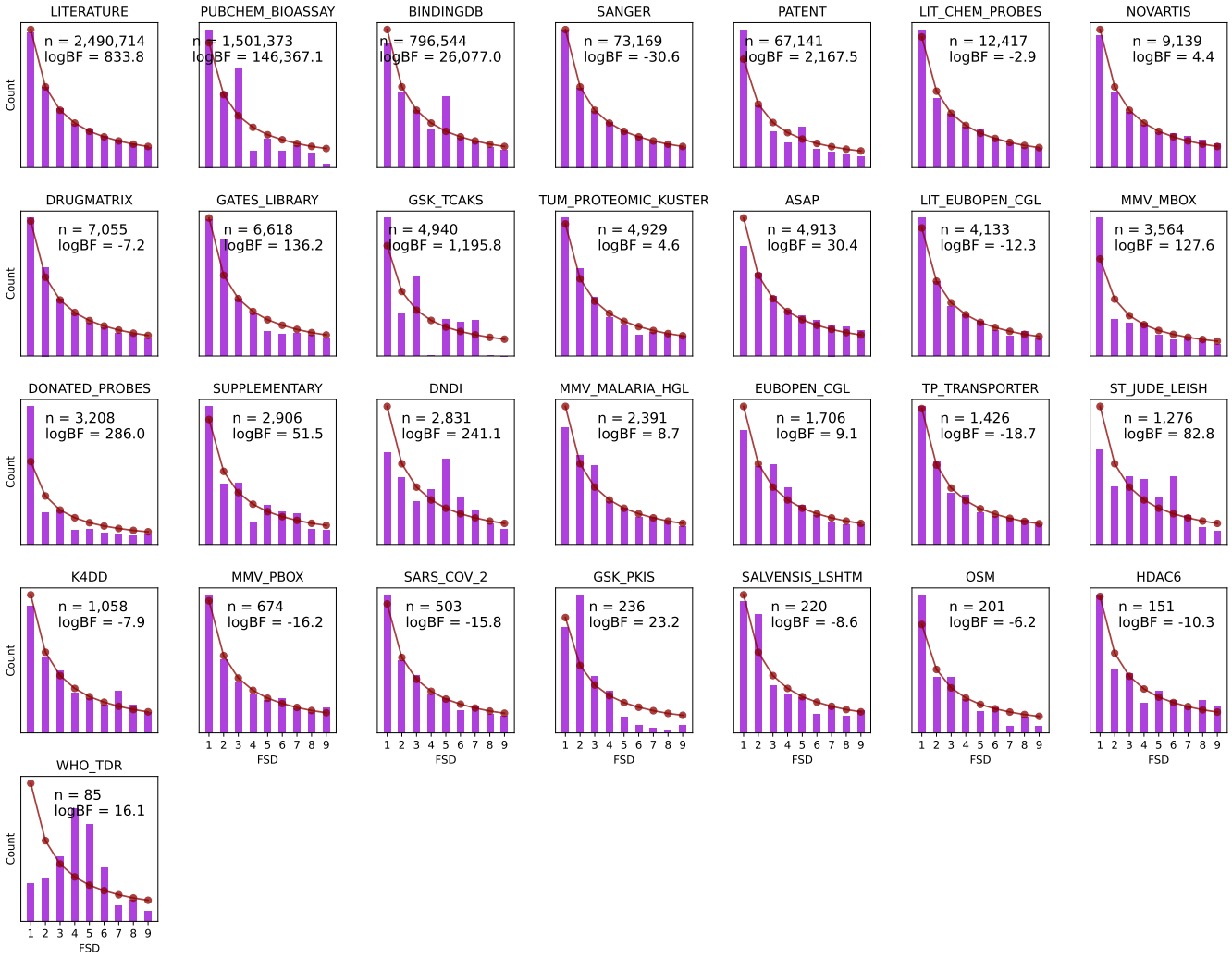


Figure S5: FSD distribution of bioactivity data in the ChEMBL database aggregated by data source for all sources with more than 80 datapoints.

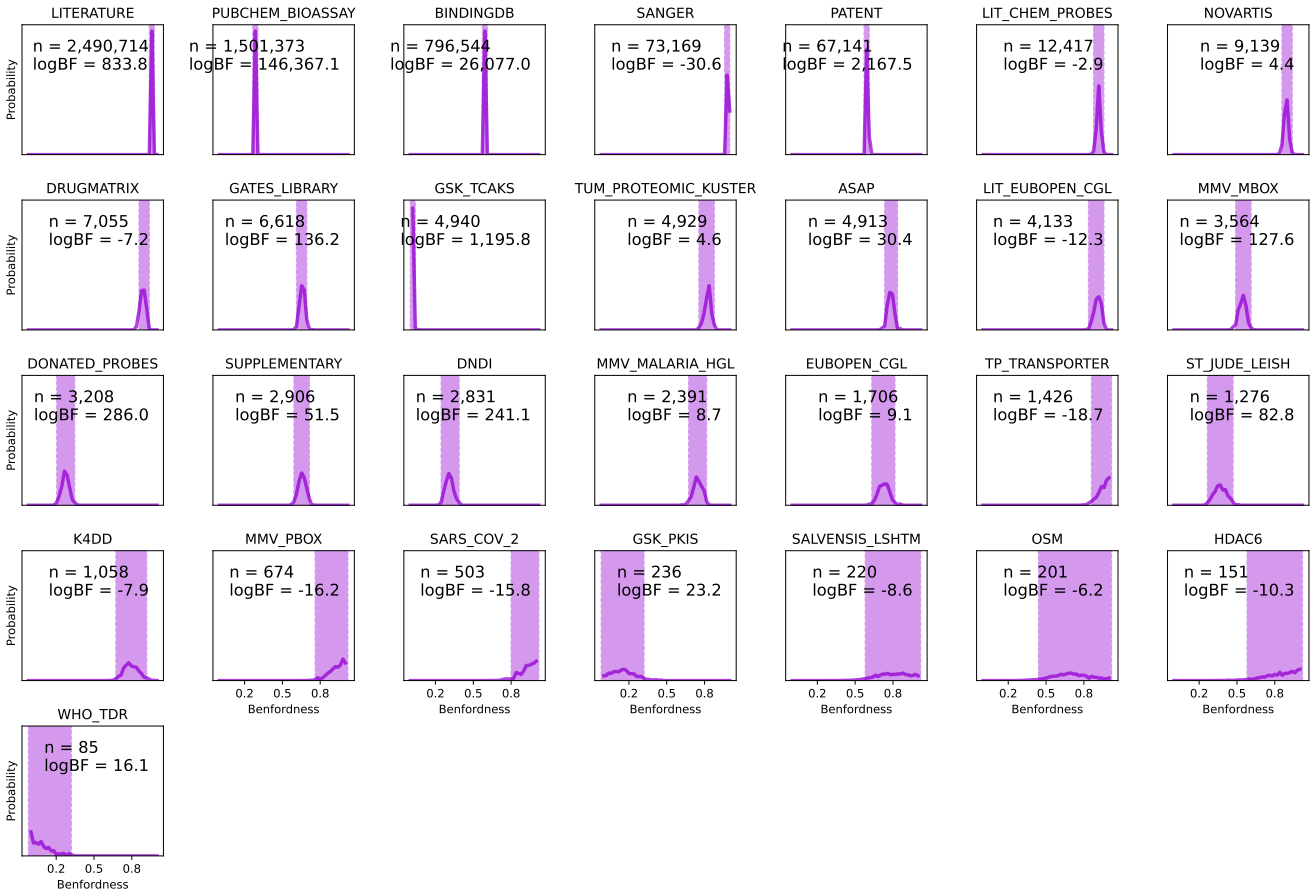


Figure S6: Estimated Benfordness by SBBE of bioactivity data in the ChEMBL database aggregated by data source for all sources with more than 80 datapoints.

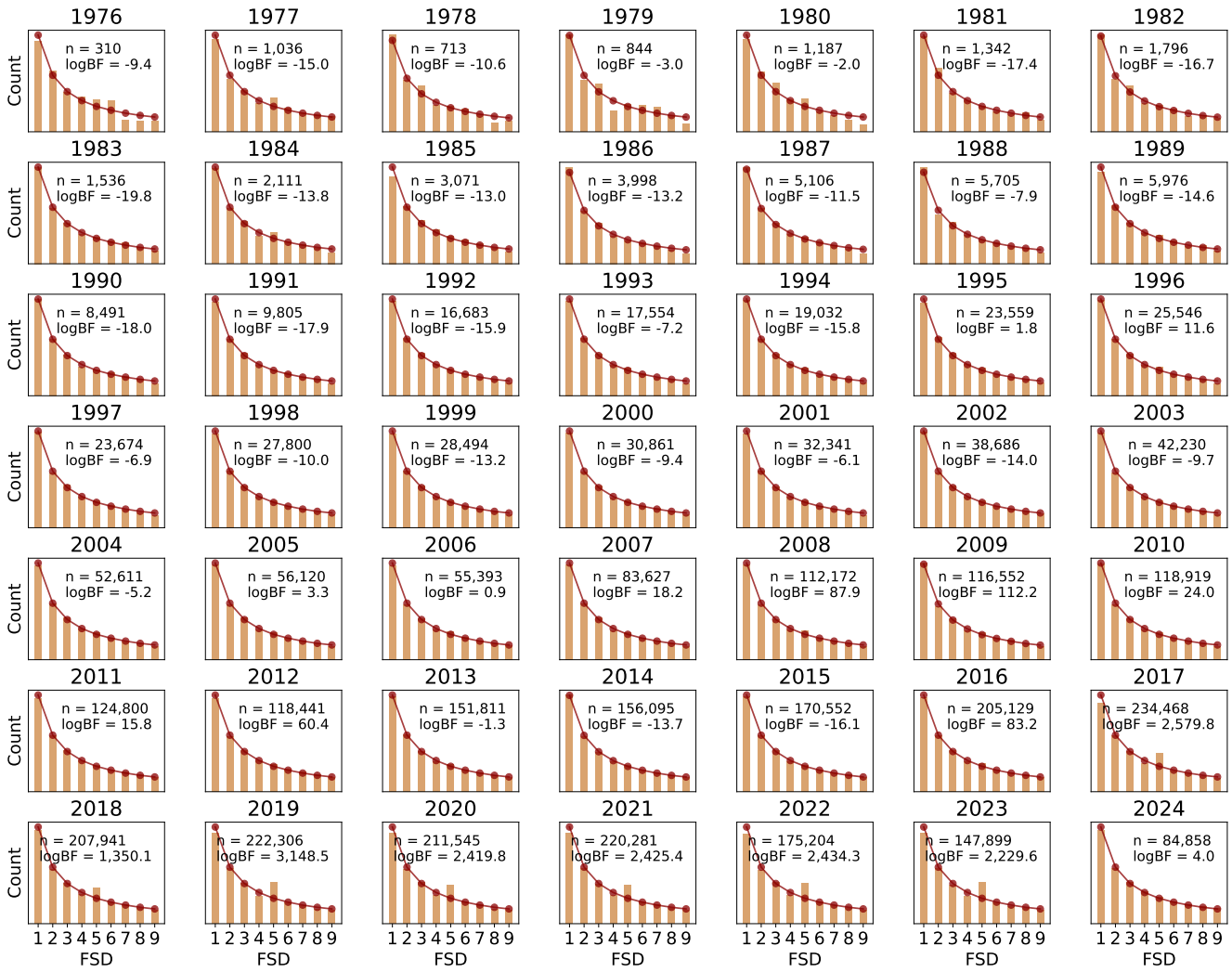


Figure S7: FSD distribution of bioactivity data in the ChEMBL database aggregated by publication year.

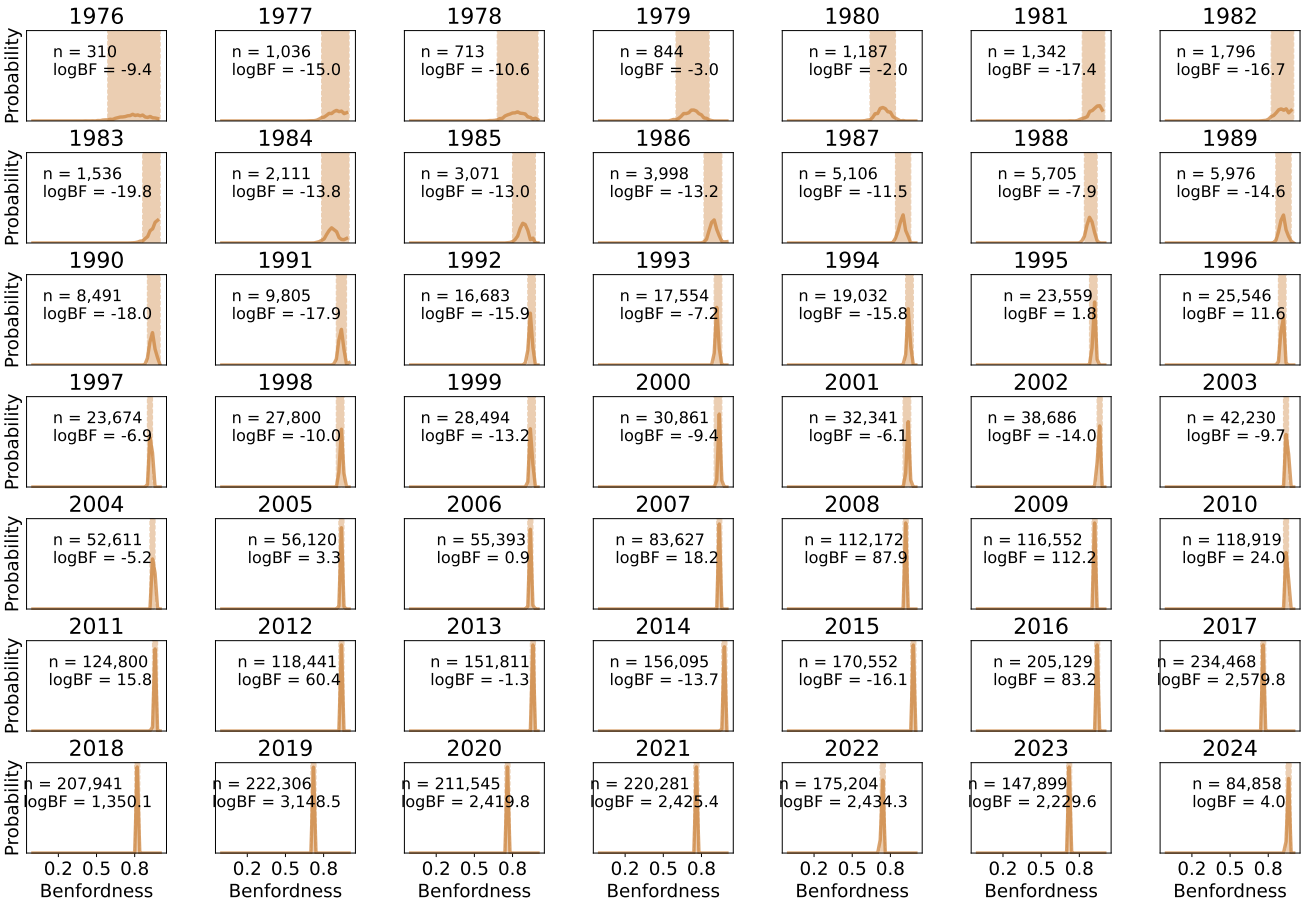


Figure S8: Estimated Benfordness by SBBE of bioactivity data in the ChEMBL database aggregated by publication year.

Supplemental Tables

Table S1: Summary statistics of categorized ChEMBL assays.

Table S1: Median of sample size (#entries), interpercentile range between 1st and 99th percentile in logarithm space ($IQR_{1\text{-}99}$), and SBBE Benfordness for ChEMBL assays (SBBE), grouped into six attributed categories, an unattributed category, and non-flagged assays with corresponding group sizes n.

Category	n	#entries	$IQR_{1\text{-}99}$	SBBE
Powers-of-ten	11	128	2.48	0.02
Log-rounding 0.05	110	936	2.55	0.02
Log-rounding 0.1	285	208	3.22	0.04
Midpoint imputation	64	163	2.55	0.00
Truncation	78	166	2.81	0.10
Repeat values	30	230	2.62	0.16
Non-flagged	3442	135	3.29	0.82
Unattributed	288	132	2.25	0.34

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